

Alpilla 600 MW Combined Cycle Power Plant

General Mechanical Specification

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Originator	Istanbul EPC Mühendislik ve Taahhüt A.Ş.
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A	IFR	2026-12-14	First issue for Owner approval	EPCE	EPC mechanical lead	EPC engineering manager

1. Purpose and scope

1.1 This specification defines the general mechanical requirements for rotating and static equipment of all balance-of-plant (BOP) mechanical packages of the 1 x 1 combined cycle power plant. It is appended to each package requisition and technical specification; package specifications may add to it and shall not relax it without a technical query approved by IEPC.

1.2 The specification does not apply to the gas turbine, steam turbine, generators and HRSG of the IEC power island, which follow the IEC contract specifications, except where this specification is referred to in the interface documents.

1.3 Where a package specification, a vendor standard and this specification conflict, the most stringent requirement applies until IEPC has resolved the conflict in writing. Deviations shall be listed in the vendor deviation list with the clause number of this specification.

2. Governing requirements

ER-01.03: All equipment and structures shall be designed for an economic life of 30 years, 8,000 equivalent operating hours per year and up to 250 starts per year (of which up to 50 cold, 100 warm and 100 hot starts), including daily cycling and load following.

ER-02.08: Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

ER-14.03: Maximum 85 dB(A) at 1 m from any equipment in normal operation.

ER-15.02: ITPs for all major equipment with hold and witness points for the Engineer, 15 days' notice for witnessed tests.

ER-15.04: Material certificates per EN 10204 3.1 (3.2 for pressure parts above 50 bar) and positive material identification of alloy piping.

ER-17.06: Commissioning spares, special tools, and a recommended two-year operating spares list with prices valid for 18 months.

3. Site and ambient design data

3.1 Equipment shall be designed for the site and ambient data below. Outdoor equipment shall be suitable for the marine C5 atmosphere of the site (see ALP-EPC-00000-ME-RPT-0002 and ALP-EPC-00000-ME-SPC-0002).

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001
Chloride deposition maximum	420 mg/m2/day	ISO 9225, category S3	confirmed	SRC-OWNR-0001, ER-02.10
Ground flash density	2.4 flashes/km2/year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001, ER-02.12

Parameter	Value	Condition	Status	Basis
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001, ER-02.11
PM10 maximum (event)	260 ug/m3	Saharan dust episodes	confirmed	SRC-OWNR-0001, ER-02.10
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001
Sea-salt aerosol annual mean	14 ug/m3	as NaCl	confirmed	SRC-OWNR-0001
Sea-salt aerosol maximum (event)	165 ug/m3	as NaCl, lodos storms	confirmed	SRC-OWNR-0001, ER-02.10
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001, ER-02.11
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001, ER-02.11
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels >= +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m2		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

Parameter	Value	Condition	Status	Basis
Cold start time to base load	180 min	maximum	confirmed	ER-04.04
Design life	30 years		confirmed	ER-01.03
Hot start time to base load	45 min	maximum	confirmed	ER-04.04
Operating hours per year	8,000 EOH/year		confirmed	ER-01.03
Starts per year	250 starts/year	50 cold, 100 warm, 100 hot	confirmed	ER-01.03
Warm start time to base load	90 min	maximum	confirmed	ER-04.04

3.2 The design life of all equipment and structures shall be the design life given in the operating data above (DP-CASE-0034), at the stated operating hours and starts per year, including daily cycling. Components with a shorter life (seals, bearings, filters, wear parts, gaskets) shall be listed in the recommended spares list with their replacement interval.

3.3 Equipment installed outdoors shall be capable of full-capability operation between the minimum and maximum ambient temperature of DP-CASE-0007 and DP-CASE-0008 and shall survive the survival temperature range of DP-CASE-0009 without damage when not in operation.

3.4 Equipment shall be designed for the seismic load of the site (DP-CASE-0002, TBDY 2018). Anchorage, lifting lugs, nozzle loads and internal parts shall be checked for the seismic acceleration in operation and for the transport and lifting load cases. Rotating equipment shall remain operable after the design earthquake, or shall be tripped by the protection system where the package specification states so.

4. Codes, standards and general design rules

4.1 Equipment shall comply with the Turkish regulations transposing the Pressure Equipment Directive 2014/68/EU and the Machinery Directive 2006/42/EC, and with the ATEX directive 2014/34/EU in classified areas. Each vendor shall provide the CE / conformity declaration and the technical file for the equipment supplied.

4.2 Design, materials, manufacture, testing and documentation shall follow the codes named in the package specification. Where a code is not named, the order of precedence is: Turkish law, ISO / EN, API / ASME, vendor standard (with IEPC approval).

4.3 Equipment shall be selected as proven design with at least 3 units of the same model in comparable service for at least 8,000 operating hours; prototypes require written approval of the Owner via IEPC.

4.4 Equipment shall be sized from the heat and mass balance cases of the project with the margins of the equipment sizing basis: pumps +10 % on flow and +5 to 10 % on head, motors to the next IEC standard size above the maximum absorbed power (including margins) of the driven equipment.

4.5 Maximum allowable noise of any equipment in normal operation shall not exceed the near-field noise limit given in the table below, measured at 1 m distance in free field. Vendors shall guarantee the sound pressure level and sound power level for each equipment item and shall provide acoustic enclosures, silencers or lagging as needed, without impairing access for maintenance.

Parameter	Value	Condition	Status	Basis
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03
Noise at nearest receptor (day)	55 dB(A)	LAeq day	confirmed	SRC-OWNR-0007, ER-14.04
Noise at nearest receptor (night)	45 dB(A)	LAeq night	confirmed	ER-14.04

5. Pumps

5.1 Centrifugal pumps in hydrocarbon (LDO), flammable, toxic or high-energy services (rated power above 150 kW or discharge pressure above 40 bar) shall comply with API 610 / ISO 13709. Pumps in water, utility and other non-hazardous services shall comply with ISO 5199 (or ISO 9908 for Class III up to 150 kW) or API 610 by package specification; vertical wet-pit circulating water pumps shall follow ANSI/HI 9.8 and the ER-08.03 requirements.

5.2 Pump selection shall provide a NPSH available margin over NPSH required (NPSH3) of at least 1.0 m or 1.3 x NPSH3 at rated flow, whichever is greater, and 1.5 m or 1.5 x NPSH3 for boiler feed and condensate pumps at minimum hotwell / deaerator level. NPSH available shall be calculated on the lowest liquid level and the highest liquid temperature of the operating cases.

5.3 Pumps shall have a continuously rising head-flow curve to shut-off, with a head rise of at least 5 % from rated point to shut-off; rated flow shall lie within 70 to 110 % of the best efficiency point (BEP) flow, and the minimum continuous stable flow shall be stated on the datasheet.

5.4 Vendors shall guarantee pump efficiency at rated point with a tolerance of 0 % on the rated flow and head (ISO 9906 Grade 2B), and the efficiency shall be not lower than the hydraulic institute (ANSI/HI 14.3) efficiency band for the duty. Pumps of rated power above 200 kW shall have the efficiency confirmed by a performance test.

5.5 Shaft sealing of pumps handling LDO, ammonia solution, hazardous or volatile liquids shall be single or dual mechanical seals to API 682 (or ISO 21049) with the piping plan selected per service; magnetic drive pumps are accepted for aqueous ammonia when the vendor demonstrates dry-run protection.

5.6 Vibration of pumps and drivers shall be within zone A / B of ISO 10816-7 (rotodynamic pumps) and ISO 20816-3 (motors and other industrial machines) at all continuous operating points; acceptance is at the factory and at site testing. Pumps of 200 kW and above shall have provision for proximity probes or accelerometers on the bearings.

5.7 Vertical CW pumps and other pumps with wetted parts in seawater shall use the materials of ALP-EPC-00000-ME-RPT-0002 (super duplex or equivalent, rubber-lined columns); dissimilar-metal contact shall be electrically isolated as stated there.

5.8 Pump casing hydrotest pressure shall be 1.5 x maximum allowable working pressure; the barrel and volute of pumps with cast casings shall have a casting factor of not less than 0.8 unless radiography or ultrasonic testing is applied.

6. Fans, blowers and compressors

6.1 Fans shall comply with ISO 5801 for performance and AMCA / ISO 14694 for vibration quality (balancing grade G 2.5 or better, BV-3 category). Fan static efficiency shall be stated at the duty point and shall not be lower than the ISO 12759 minimum efficiency grade applicable to the fan category. Impellers of aluminium or GRP in outdoor use shall be coated and electrically isolated from steel structures in accordance with ALP-EPC-00000-ME-RPT-0002.

6.2 Compressors (instrument and service air, nitrogen) shall be oil-free (screw or centrifugal) for instrument air; performance shall be tested per ISO 1217, with a specific power guarantee at rated pressure. Air receivers shall be pressure vessels according to the vessel section of this specification.

6.3 Rotating equipment with rated power above 200 kW shall have a lateral and torsional analysis (where the driver is variable speed or the shaft train has more than one coupling) and shall be balanced to ISO 21940-11 grade G 2.5 (G 1 for high-speed machines above 3,600 rpm). The vibration limit shall be the alarm and trip set points given in ISO 20816.

7. Pressure vessels, tanks and heat exchangers

7.1 Pressure equipment shall be designed, manufactured and certified per the Pressure Equipment Directive 2014/68/EU (conformity module and category per package specification) with EN 13445 or ASME Section VIII Div. 1 as design code; the code shall be stated on the datasheet and shall not be mixed within a vessel. Vessels with design pressure above 0.5 bar(g) but outside the scope of the directive shall be designed to the same codes.

7.2 Vessel design pressure and design temperature shall be set from the maximum operating conditions of the process cases plus a margin: design pressure at least 10 % (or 1.5 bar, whichever is greater) above the maximum operating pressure or the relief valve set pressure, and design temperature at least 15 K above the maximum operating temperature; vacuum service shall be full vacuum. The values shall agree with the equipment register (design_pressure, design_temperature).

7.3 Corrosion allowance of vessels, tanks and heat exchangers shall be as stated in ALP-EPC-00000-ME-RPT-0002; minimum nominal plate thickness of carbon steel shells shall be 6 mm (4 mm for stainless steel), excluding corrosion allowance.

7.4 Atmospheric storage tanks for LDO and fire water shall comply with API 650 (fixed cone roof; internal floating roof not required for LDO with the flash point of DP-EPCE-0013), with foundation ring wall and leak detection under the bottom. Tank bottoms shall have a slope and a sump for water draw-off. Tanks handling flammable liquids shall be designed with NFPA 30 spacing, venting and bund requirements. Shop-built tanks of demineralised water and chemicals shall comply with EN 14015 or the package standard (API 650 for steel tanks above 10 m³ in flammable service).

7.5 Storage tanks shall be designed with a wind speed and seismic load as per the site data (DP-CASE-0004, DP-CASE-0002), with the sloshing analysis for the seismic case (API 650 Annex E or equivalent) and with settlement tolerance as per the civil design.

7.6 Shell-and-tube heat exchangers shall comply with TEMA (class C or R by service) and the pressure vessel code. Plate heat exchangers shall have 10 % surface margin over the calculated duty with fouling factors per the datasheet and shall provide space for tube or plate removal. Seawater-cooled exchangers shall use titanium or higher-alloy plates or tubes; the fouling / cleanliness basis shall be the cleanliness factor of the condenser design where applicable.

7.7 Vessels shall have manholes (minimum DN 500, DN 600 for vessels above 2 m diameter), lifting lugs or trunnions designed for 1.5 x the empty weight (dynamic factor 2.0 for lifting of the full package), drain and vent connections at the lowest and highest points and earthing bosses (2 per vessel, on opposite sides).

7.8 Welding of pressure equipment and its non-destructive examination shall follow ALP-EPC-00000-PI-SPC-0003 and ASME Section IX or EN ISO 15614 as qualified by the manufacturer; welders and procedures shall be approved by an inspection body.

8. Electric motors and drivers

8.1 Electric motors shall comply with IEC 60034, with insulation class F and temperature rise class B at the maximum ambient temperature of the site; LV motors of 0.75 kW and above efficiency class IE3 minimum (IEC 60034-30-1), IE4 for continuously running LV motors above 75 kW; MV motors with the guaranteed efficiency on the datasheet (EL-DBR-0001 section 8).

8.2 The supply voltage of motors shall follow the auxiliary electrical system concept: motors of 200 kW and above are supplied from the 10.5 kV medium-voltage system; motors below 200 kW from the 0.42 kV low-voltage system. The sizing basis and the voltage threshold shall not be deviated from without a change of the load list and of the decision.

8.3 MV motors shall be designed for the short-circuit level of the MV switchgear and for direct on-line starting within the voltage dip and terminal voltage limits of the electrical design criteria (EL-DBR-0001 section 8). Larger motors shall be started through a variable frequency drive or a soft-starter where the starting study requires it.

8.4 Motors for outdoor installation shall have degree of protection IP55 as a minimum (IP56 for motors in the CW pump house, IP23 with heat exchange only in indoor dry rooms), corrosion protection per ISO 12944 C5 in accordance with ALP-EPC-00000-ME-SPC-0002 and anti-condensation heaters for motors above 15 kW.

8.5 Motors above 200 kW shall have winding temperature sensors (2 per phase, PT100), bearing temperature sensors and vibration monitoring provisions; motors in classified areas shall be Ex certified (Ex d, Ex e or Ex p) for the zone and gas group per the hazardous area classification.

8.6 Diesel-engine driven equipment (fire pump, emergency generator) shall comply with the package specification and NFPA 20 (fire pump); the engine ratings shall be given at the site altitude and the maximum ambient temperature.

9. Nameplates, tagging and marking

9.1 Every equipment item shall be tagged with the KKS number of the project equipment register (VGB-S-811 and the project key list) on a stainless steel nameplate (316L, 1 mm minimum) fixed with stainless rivets or screws. The nameplate shall show: KKS tag, manufacturer, type and serial number, year of manufacture, design conditions, weight, and the CE / PED mark where applicable.

9.2 Field instrument, valve and small equipment tags shall be identified by a stainless steel tag plate with the KKS number, wired to the equipment; equipment shall be identifiable from the ground.

9.3 Nameplates, rotation arrows, lubricant identification and safety signs shall be in English and Turkish where required by Turkish regulations.

10. Lifting, handling and maintainability

10.1 Every equipment item and every component heavier than 25 kg that must be handled for maintenance shall have lifting lugs, eyebolts or lifting points designed to EN 13155 (or the applicable code) for at least 2.0 x the working load; the lifting arrangement shall be shown on the general arrangement drawing. Items above 500 kg shall have a lifting plan and a designated lifting device (crane, monorail, or davit) in the layout.

10.2 Equipment shall be arranged with clear maintenance envelopes to allow removal of the largest component (motor, pump rotor, tube bundle, filter element) without dismantling of other equipment or piping; the envelope shall be shown in the layout. Vendors shall state the heaviest maintenance item and its removal path.

10.3 Guards shall cover all exposed rotating parts and hot surfaces above 60 degC that can be touched; guards shall be of the removable type without special tools, and shall meet the Machinery Directive.

11. Painting, insulation and preservation

11.1 Painting and coating of equipment shall follow ALP-EPC-00000-ME-SPC-0002; the insulation of hot equipment shall follow ALP-EPC-00000-ME-SPC-0003. Equipment delivered pre-painted shall have the coating system and DFT recorded in the ITP dossier.

11.2 Equipment shall be preserved for shipment, storage on site for 12 months and up to 24 months where erection is delayed, in accordance with the vendor preservation procedure approved by IEPC: internal surfaces protected against corrosion, rotating machines rotated periodically, desiccant with humidity indicators for enclosed spaces, and all openings blanked or capped. Preservation records shall be maintained and handed over.

11.3 Long-term storage and lay-up of equipment after installation shall follow the manufacturer's lay-up instructions; the dry and wet lay-up requirements for the HRSG and steam cycle equipment of ER-06.08 shall be observed.

12. Inspection, testing and documentation

12.1 Vendors shall issue an inspection and test plan (ITP) for each major equipment item with hold points (H), witness points (W) and review points (R); the Engineer shall be given 15 days' notice of witnessed tests. Manufacturing shall not proceed beyond a hold point without release.

12.2 Material certificates shall be EN 10204 type 3.1 for all pressure-retaining and load-carrying parts; type 3.2 for pressure parts with design pressure above 50 bar; positive material identification (PMI) shall be carried out on 100 % of alloy components and alloy piping of the package.

12.3 Factory acceptance test shall include as a minimum: hydrostatic test of pressure parts, performance test (pumps of rated power above 200 kW at full speed), mechanical run test with vibration and bearing temperature recording, noise measurement for machines above 200 kW, and leak / functional tests of skids. Test procedures shall be approved before the test.

12.4 Vendors shall provide with each equipment item: certified drawings, datasheets in the format of the project, material certificates, test reports, O&M manual, list of commissioning and two-year operating spares with prices valid for 18 months, and special tools.

12.5 Spare parts: vendors shall supply commissioning spares with the equipment, and a recommended two-year operating spares list. A common spare pump assembly shall be provided for the CW pumps as stated in the governing requirement; further spare units are defined in the package specification.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Work Equipment Regulation	Regulation on Health and Safety in the Use of Work Equipment
TBDY 2018	Turkish Building Earthquake Code (Turkiye Bina Deprem Yonetmeligi)
PED 2014/68/EU	Pressure Equipment Directive, transposed as the Pressure Equipment Regulation (Basincli Ekipmanlar Yonetmeligi)
Machinery Directive 2006/42/EC	Machinery Directive, transposed as the Machinery Safety Regulation (Makine Emniyeti Yonetmeligi)
ATEX 2014/34/EU	ATEX Equipment Directive, transposed as the regulation on equipment for potentially explosive atmospheres
IEC 60034-1	Rotating electrical machines - Part 1: Rating and performance
ASME Section VIII Div. 1	ASME BPVC Section VIII Division 1 - Pressure Vessels
ASME Section IX	ASME BPVC Section IX - Welding, Brazing and Fusing Qualifications
EN ISO 15614	Specification and qualification of welding procedures for metallic materials - Welding procedure test
EN 10204	Metallic products - Types of inspection documents
API 650	Welded Tanks for Oil Storage
NFPA 30	Flammable and Combustible Liquids Code
NFPA 20	Standard for the Installation of Stationary Pumps for Fire Protection
EN 13001	Cranes - General design (all parts)
ISO 20816-2	Mechanical vibration - Measurement and evaluation of machine vibration - Part 2: Land-based gas turbines, steam turbines and generators above 40 MW with fluid-film bearings